GOVERNMENT OF WESTERN AUSTRALIA

GSWA Geochronology Record

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Department of Mines, Industry Regulation and Safety

211111.1.pdf *Geochronology Record 1910*

211111.1: meta-andesite, McCrae Bore

(Madoonga Formation; Southern Cross Supergroup, Youanmi Terrane, Yilgarn Craton)

SS Romano, IOH Fielding, MTD Wingate, Y Lu and TJ Ivanic

Location and sampling

BELELE SG50-11, KALLI 2344 Zone 50, MGA 537838E 7013837N
Warox Site TJIMKT000131 Sampled on 8 September 2014

This sample was collected from a low-lying outcrop on the north side of Weld Range, about 10.3 km south-southwest of Mardoonganna Hills, 7.8 km northeast of Errida Spring and 3.5 km west-northwest of McCrae Bore.

Geological context

The unit sampled is a meta-andesite with lenticular compositional heterogeneity assigned to the Madoonga Formation of the 3010–2920 Ma Southern Cross Supergroup (Ivanic et al., 2022). The Southern Cross Supergroup consists of metamorphosed basalt, andesite–dacite–rhyolite, and variable proportions of sedimentary rocks exposed in several belts along a northwest-trending, 750 km-long corridor from Ravensthorpe in the south to the Weld Range in the north within the Youanmi Terrane (Ivanic et al., 2022). This supergroup contains six formations, i.e., the Wongan, Madoonga, Gossan Hill, Mount Gibson, Honman, and Annabelle Volcanics Formations, which are unconformably overlain by the c. 2.8 Ga Norie and Polelle Groups of the Murchison Supergroup (Van Kranendonk et al., 2013; Ivanic et al., 2022). The 2979–2969 Ma Madoonga Formation includes metamorphosed rhyolitic volcanic and volcaniclastic rocks with lesser psammite and pelite and minor komatiitic basalt (Wyche et al., 2014). A sample of metarhyolite of the Madoonga Formation located about 9.4 km to the east yielded an igneous crystallization age of 2977 ± 3 Ma (GSWA 155569, Wingate et al., 2019). A metasandstone located about 13.9 km to the east indicates a maximum depositional age of 2969 ± 3 Ma (GSWA 184112, Wingate et al., 2008).

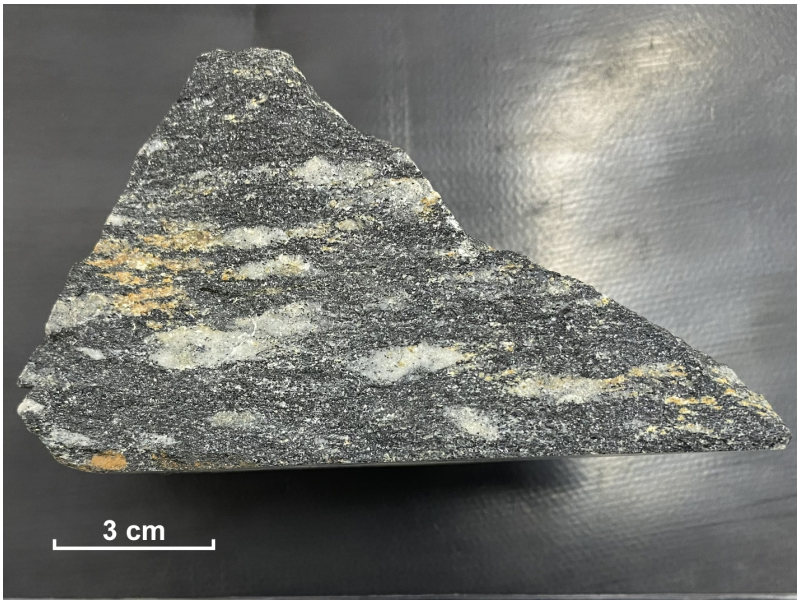

Figure 1. Hand specimen image for 211111: meta-andesite, McCrae Bore

Petrographic description

The sample is a recrystallized amphibolite characterized by a lenticular compositional heterogeneity (Fig. 1), consisting of about 30–35% hornblende, 27–30% sericitised plagioclase, 20–25% aggregates of fine-grained quartz and unaltered plagioclase, 8–10% microcline, 7–10% altered pyroxene and accessory zircon and titanite. Hornblende is euhedral to subhedral, <0.05 mm in diameter, and form poorly-defined bands and lenses of non-aligned grains. The hornblende-rich

zones also contain fine-grained sericitised minerals (most likely completely altered plagioclase), and lesser amount of fine-grained quartz. Former pyroxene, which occurs as elongate grains up to 0.5 mm long, is completely replaced by epidote and fine-grained brown iron oxides. Hornblende-poor lenses comprise recrystallized aggregates of quartz, plagioclase and K-feldspar >0.1 mm in size, and anhedral and micropoikilitic grains of microcline and rare plagioclase <2.5 mm in length. The rock has been completely recrystallized to a granophyric and partly compositionally segregated texture at amphibolite facies. The presence of metamorphic amphibole in this rock is interpreted to be of secondary importance to the protolith being intermediate in composition, as indicated by the presence of 20% primary magmatic quartz.

Zircon morphology

Zircons isolated from this sample are colourless and mainly subhedral. The crystals are up to 150 µm long, and equant to elongate with aspect ratios up to 5:1. In cathodoluminescence (CL) images, most zircons exhibit concentric zoning. A CL image of representative zircons is shown in Figure 2.

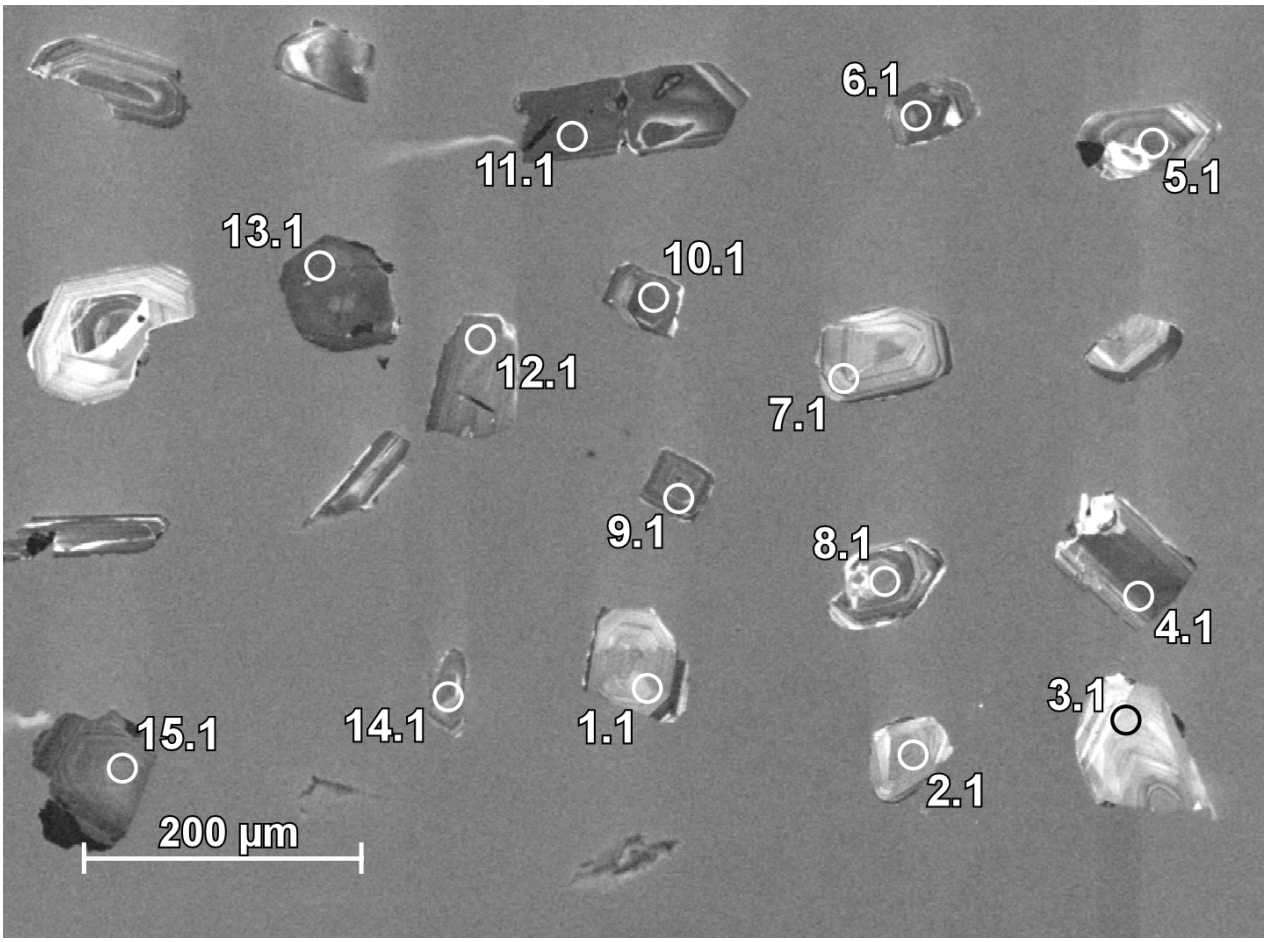

Figure 2. Cathodoluminescence image of representative zircons from 211111: meta-andesite, McCrae Bore. Numbered circles indicate the approximate locations of analysis sites

Analytical details

This sample was analysed on 21–22 October 2021, using SHRIMP-B. Fifteen analyses of the M257 standard were obtained during the session, of which 11 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.80% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.32% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1 (provided in the Links section). Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

A total of 15 analyses were obtained from 15 zircons. Results are listed in Table 1 and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to slightly discordant (Fig. 3), and form a single group, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 15 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2977 ± 5 Ma (MSWD = 0.87).

The date of 2977 ± 5 Ma for the 15 analyses in Group I is interpreted as the igneous crystallization age of the meta-andesite.

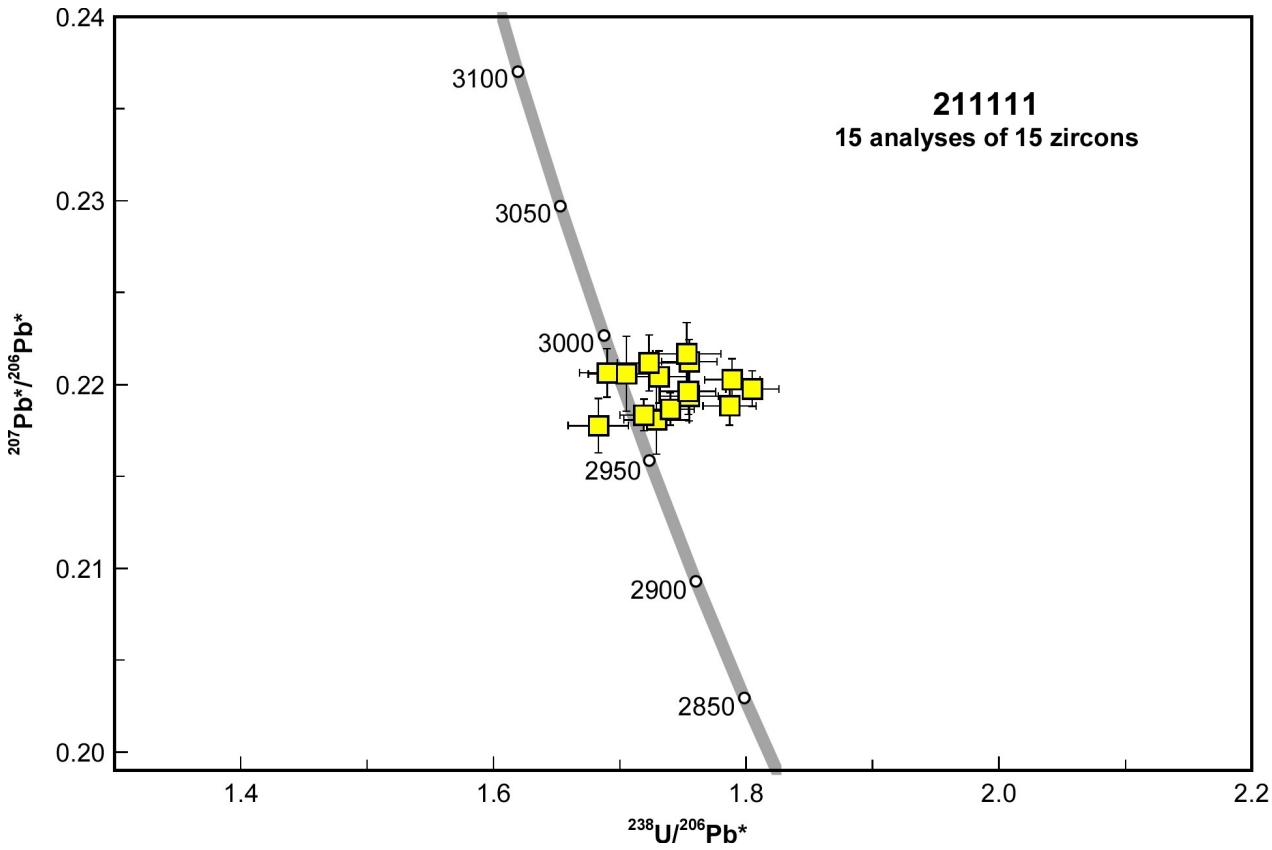

Figure 3. U–Pb analytical data for zircons from sample 211111: meta-andesite, McCrae Bore. Yellow squares indicate Group I (magmatic zircons)

Links

Table 1. Ion microprobe analytical results for zircons from sample 211111: meta-andesite, McCrae Bore: 211111.1.Table1.xls
SHRIMP spot images: 211111_spots.pdf
Introduction to geochronology information: Intro_2023.pdf

References

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Van Kranendonk, MJ, Ivanic, TJ, Wingate, MTD, Kirkland, CL and Wyche, S 2013, Long-lived, autochthonous development of the Archean Murchison Domain, and implications for Yilgarn Craton tectonics: Precambrian Research, v. 229, p. 49–92.

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Recommended reference for this publication

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Grid references in this publication refer to the Geocentric Datum of Australia 2020 (GDA20). All locations are quoted to at least the nearest 100 m.

Capitalized names in text refer to standard 1:100 000 map sheets, unless otherwise indicated.

WAROX is GSWA's field observation and sample database. WAROX site IDs have the format 'ABCXXXnnnnnnSS', where ABC = geologist username, XXX = project or map code, nnnnnn = 6-digit site number, and SS = optional alphabetic suffix (maximum 2 characters).

Isotope and element analyses are routinely conducted using the GeoHistory laser ablation ICP-MS and Sensitive High-Resolution Ion Microprobe (SHRIMP) facilities at the John de Laeter Centre (JdLC), Curtin University, with the financial support of the Australian Research Council and AuScope National Collaborative Research Infrastructure Strategy (NCRIS), and the Microscopy Australia laser ablation ICP-MS facility at the Centre for Microscopy, Characterisation & Analysis (CMCA), The University of Western Australia (UWA). The TESCAN Integrated Mineral Analyser (TIMA) instrument was funded by a grant from the Australian Research Council (LE140100150) and is operated by the JdLC with the support of the Geological Survey of Western Australia, UWA and Murdoch University. Mineral analyses are routinely obtained using the electron probe microanalyser (EPMA) facilities at the CMCA at UWA, at Adelaide Microscopy at the University of Adelaide and the Central Science Laboratory at the University of Tasmania.

Digital data related to WA Geology Online, including geochronology and digital geology, are available online at the Department's Data and Software Centre and may be viewed in map context at GeoVIEW.WA. For further information about this Geochronology Record, please contact geochronology@dmirs.wa.gov.au and quote the geochronology record and/or the sample number.

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Department of Mines, Industry Regulation and Safety

100 Plain Street

EAST PERTH, WA 6004

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